Eyes open / eyes closed

Your first Nexus experiment - a practical guide for a two-student team.

> **Your question:** Can a classifier distinguish eyes-open and eyes-closed trials in a new recording from the same participant, better than a 50% balanced-chance benchmark?

1. Agree the plan before collecting data

One participant, one operator. The participant sits comfortably; the operator runs cues, watches for mistakes and writes notes. Use a participant code such as **P01**, not a name. Arrange permission with your teacher. If you swap roles, the second person is P02 and needs their own training/test pair.

Recording	Name in Nexus	Initial target
Training	P01-training-01	20 accepted Eyes open + 20 accepted Eyes closed trials
Final test	P01-test-01	A fresh recording: 20 accepted trials of each task

These are pilot targets, not a guarantee of a strong result. Take breaks. Use a few synthetic trials to learn the controls first; synthetic data cannot be your primary evidence.

Write down the electrode locations, trial length, rest time, room lighting and exclusion rules. For example: exclude an unclear/mixed eye state, a missed cue, obvious movement/blink artefact, interruption or failed signal quality. Use the same rules for both recordings.

2. Set up a steady signal

Follow the OpenBCI headset and EEG setup instructions with your teacher/adult helper. In **Console**, connect the correct board and click **Start**. Check advancing samples and electrode/reference contacts. Set your **local mains frequency**: **50 Hz in the UK**, or **60 Hz where applicable**. Keep it the same for training and test. Finish any earlier practice recording before starting the lab session.

The eye preset suggests **O1 and O2**, at the back of the head. Confirm that the channel labels match the actual electrode positions. Do not assume the app's fallback selection is appropriate for real eye-state data. Keep the same selected channels and their order in training and test.

3. Start the training recording

Open **Experiment Lab**. Enter participant **P01**, recording name **P01-training-01** and task preset **Eyes open / closed**. Select your agreed electrodes. Click **Start labelled recording**.

Set **Trial length** to **8 seconds** and decide whether to enable **Speak cue**. Keep these choices for the test. Wait for **3/3 clean windows** and the enabled **Record next trial** button. If it stays disabled, improve signal quality; do not weaken the rules to make it start.

Collect and label each trial

The app records the cue label automatically. The operator checks what actually happened.

4. Follow each cue

Click **Record next trial**. Nexus chooses among the tasks with the fewest completed included trials, with a random choice when tied. There is a **3-second Get ready** cue before the 8-second recording.

Eyes open: look softly at the same plain spot; do not read or use a phone.
Eyes closed: close gently, without squeezing.
In both tasks, stay still, relax the jaw and follow the cue until the trial finishes. The operator handles the computer. Between trials, reopen the eyes, rest for an agreed interval (for example 5-10 seconds), and write any observations.

Keep the Experiment Lab tab visible during a trial. Switching views or hiding the tab interrupts and excludes it. If you need to stop early, use **Stop & exclude trial**. If an incomplete trial is left after a refresh, use **Exclude interrupted trial**.

5. Save and review labels before looking at scores

Watch the included counts for both labels and aim for the agreed target. Once the current trial has finished, click the Lab's **Save recording**. In **Review labels**, choose the training recording. Use the task-label dropdown, **Include** checkbox and note field; click **Save note** after writing a note. Label/inclusion edits are saved when changed.

Check labels against the observed eye state. If the whole trial clearly used the wrong state, correct its label and explain why. If the state changed part-way or is uncertain, exclude the trial. Never relabel from the classifier's prediction or remove a trial because it lowers the score. Automated quality failures/incomplete trials cannot simply be forced back in.

Observed event	Action before evaluation	Example note
Cue followed throughout	Keep assigned label; Include stays checked	Eyes closed throughout; still
Cue said open; eyes were closed throughout	Correct to Eyes closed; save a note	Operator observed closed eyes for whole trial
Opened midway, moved, or state uncertain	Uncheck Include; save a note	Opened eyes halfway through

Use the actual trial rows as the event log. Review all trials, not only ones that look unusual. Exclusion keeps the raw EEG; it does not delete it.

6. Collect a separate test recording

After saving training, take a break; a later day is a stronger test. Start a **new labelled recording** named **P01-test-01**. Use the same P01 participant code, board, sample rate, electrode list/order, task labels, trial length and selected local mains frequency. Repeat steps 4-5 with fresh trials.

Keep the split real: do not rename/copy the training file to make a test, reuse its trials, or randomly split nearby time windows from one recording. Review test labels using the same pre-agreed rules and your observations, before viewing predictions. Keep P02 recordings out of P01's pair.

Test once, then explain the result

Training data fits the classifier. The separate test recording measures how well it carries over.

TRAINING: P01-training-01	FINAL TEST: P01-test-01
Fits covariance reference and logistic regression. Use practice/training to settle the method.	Fresh samples from the same participant. Not used to fit or choose the method.

7. Run the evaluation

In **4. Train & test**, choose **P01-training-01** under **Training recording** and **P01-test-01** under **Test recording**. Confirm they are two different saved sessions. Click **Train pyRiemann & evaluate**.

All model fitting uses training data only. Offline checks may reject additional trials, so evaluation counts can be lower than the live included counts.

If evaluation refuses to run, read the message. Both sessions must be saved and compatible. After all quality checks, the software needs at least **5 training and 2 test trials per label**; these are operating minimums, not your study target. Missing/poor data needs more collection, not relaxed exclusions.

8. Read the result fairly

Balanced accuracy averages the fraction correct for each eye state. For two labels, the balanced-chance reference is **50%**. The confusion matrix has **actual labels in rows** and **predicted labels in columns**. Check both labels, accepted counts and rejection reasons; do not report only the best number.

Above 50% in a small test does not prove reliable brain decoding. Trials can be correlated and eye/muscle artefacts can contribute. Conclusions apply to this participant and these sessions, not other people or a measure of focus/relaxation.

After seeing the final test score, do not tune against that recording. If you change channels, labels, features or performance-based exclusions, the old test becomes development data. Collect a new untouched test recording for the final result.

9. Save the evidence

For each session, use **Export labels** (JSON) and **Download raw EEG CSV**. Keep the matching recording metadata too if backing up the app's recording folder. Use **Export result & method** for the evaluation report. Store training, test and result together under participant codes; keep participant data out of public GitHub.

Project record	Fill this in
Training name / accepted open + closed	_____
Test name / accepted open + closed	_____
Balanced accuracy / exclusions and reasons	_____
What limits your conclusion?	_____

More detail: Nexus Experiment Lab method and OpenBCI EEG / alpha demonstration. Eyes-closed posterior alpha is a useful starting hypothesis, not a guaranteed individual response.